

Q.P. Code : MCA24B208

St. Philomena's College (Autonomous), Mysore
II Semester MCA – C1 Internal Assessment Test : August - 2025
Subject : Web Technologies

Time: 75 Mins

Max Marks: 30

PART - A

Answer the following :

2 × 10 = 20

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|----|----|--|-----|---|
| 1. | a. | Explain the CSS Box Model with a neat diagram. | CO1 | 6 |
| | b. | Explain the working of a web browser and a web server with examples. | CO1 | 4 |

OR

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|----|----|---|-----|---|
| 2. | a. | Write HTML5 code to insert an image with alternate text and title. | CO1 | 6 |
| | b. | List and explain any four CSS text properties. | CO1 | 4 |
| 3. | a. | Explain DOM-based XML processing in JavaScript with a suitable example | CO2 | 6 |
| | b. | Write the syntax and example for accessing HTML elements by id, class, and tagName in JavaScript. | CO2 | 4 |

OR

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|----|----|---|-----|---|
| 4. | a. | Write a JavaScript code to validate if an email input is in the correct format. | CO2 | 6 |
| | b. | Write short notes on XPath expressions in XML. | CO2 | 4 |

PART - B

Answer the following :

2 × 5 = 10

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|----|--|--|-----|---|
| 5. | | Explain different levels of CSS with examples. | CO1 | 5 |
|----|--|--|-----|---|

OR

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|----|--|---|-----|---|
| 6. | | Write HTML5 code to create a registration form with Name, Email, Course, and Submit button. | CO1 | 5 |
| 7. | | Explain different types of events in JavaScript with suitable examples. | CO2 | 5 |

OR

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|----|--|--|-----|---|
| 8. | | Write a JavaScript program that takes a number as input and checks if it is prime. | CO2 | 5 |
|----|--|--|-----|---|

St. Philomena's College (Autonomous), Mysore
II Semester MCA – C1 Internal Assessment Test : August - 2025
Subject : Advanced Database Management Systems

Time: 75 Mins

Max Marks: 30

PART - A

Answer the following :

2 × 10 = 20

1. a. Explain the 3-tier ANSI / SPARC DBMS architecture with a neat diagram.
Also, differentiate between schema and instance.

CO1 6

- b. Discuss any four characteristics of a DBMS.

CO1 4

OR

2. a. Explain the structure and functioning of Client-Server DBMS architecture.
How is it different from a centralized DBMS?

CO1 6

- b. Write short notes on:

i) Data Independence and ii) Cloud-based DBMS

CO1 4

3. a. Draw an ER diagram for a university system that includes entities like Student, Course, Instructor, and Department. Show cardinality and relationship types clearly.

CO2 6

- b. Define Referential Integrity. Why is it important in relational database design?

CO2 4

OR

4. a. Convert the following ER components to relational schema

Entity: Student(SID, Name, Dept)

Entity: Course(CID, Title)

Relationship: Enrolled(SID, CID, Date)

Show the resulting relations with primary and foreign keys.

CO2 6

- b. What are the different types of keys in a relational database? Explain each with suitable examples.

CO2 4

PTO

Q.P. Code : MCA24B209

St. Philomena's College (Autonomous), Mysore

II Semester MCA – C1 Internal Assessment Test : August - 2025

Subject : Communication Skills and Soft Skills

Time: 75 Mins

Max Marks: 30

PART - A

Answer the following :

2 × 10 = 20

1. a. What is Encoding in communication? Give examples.
- b. Briefly explain Interpersonal communication.

CO1 05

CO1 05

OR

2. a. What are media in communication process? Explain with examples.
- b. Discuss Non-verbal communication.

CO1 05

CO1 05

3. a. What are the functions of communication? Explain.

CO2 10

OR

4. a. Give the meaning of Interview. How do you prepare yourself as an Interviewer?

CO2 10

PART - B

Answer the following :

2 × 5 = 10

5. What is Art. 19 (1)(a) of Indian Constitution? Describe.

CO1 05

OR

6. List the limitations mentioned under Art. 19(2) of Indian Constitution.

CO1 05

7. Discuss Law of Defamation in India.

CO2 05

OR

8. Write a note on Indian Copyright Act.

CO2 05

St. Philomena's College (Autonomous), Mysore
II Semester MCA – C1 Internal Assessment Test : August - 2025
Subject : Java Programming

Time: 75 Mins

Max Marks: 30

PART - A

Answer the following :

2 × 10 = 20

1. a. Explain Automatic conversions and Casting Incompatible type.
- b. Justify the Statement "Compile once run anywhere" in JAVA.

CO1 6

CO1 4

OR

2. a. Interpret the general form of a class with example.
- b. Write a Java program to sort the elements using a for loop.

CO1 4

CO1 6

3. a. Discuss the uses of super keyword with example.
- b. Explain method overriding with a suitable example.

CO2 4

CO2 6

OR

4. a. What is abstract class and abstract method? Explain with an example
- b. Define package. How to import a package in java?

CO2 6

CO2 4

PART - B

Answer the following :

2 × 5 = 10

5. Explain the concept of passing object as parameters with an example.

CO1 5

OR

6. Develop the java program to create a class 'Employee' which contains 'name', 'designation', 'empid', 'basic salary', as variable and read() and write () as method. Using this class read and write an employee information from main() method.

CO1 5

7. Illustrate with an example a superclass variable can reference to a subclass object.

CO2 5

OR

8. Write a Java program to implement multilevel inheritance with 3 levels of hierarchy.

CO2 5
